

DIFFERENTIAL EQUATIONS

From Local Change to Evolution

Calculus, Dynamics, and Waves

Extended mathematical problem

Part I

Extended paper — intended for several sittings

Calculators and electronic aids are not required.

Adapted from the *Differential Equations* notes based on lectures by M. G. Worster. Original notes by Dexter Chua.

Instructions and planned structure

The completed paper will consist of one problem divided into the following ten parts. This instalment contains Part I.

- I. Differentiation, asymptotics, and local approximation;
- II. integration and multivariable differentiation;
- III. first-order linear differential equations;
- IV. nonlinear first-order equations and discrete dynamics;
- V. homogeneous second-order equations and phase space;
- VI. forced and equidimensional equations, recurrences, damping, and impulses;
- VII. series solutions near ordinary and singular points;
- VIII. gradients, Taylor forms, and stationary points;
- IX. differential systems and phase portraits;
- X. characteristics, waves, and diffusion.

Question numbers will remain continuous when later parts are added. The result of any question may be used later, even if that question has not been answered, provided its number is cited clearly. Indications are optional. Definitions and notation are introduced when they first become necessary. No result may be used unless it belongs to the calculus background specified below, has been stated as an admitted result, or has been proved in an earlier question.

Part I. — Differentiation, asymptotics, and local approximation

As in the original notes, the reader is assumed to know elementary single-variable real calculus: real limits and their algebra, continuity, the usual elementary functions and their derivatives, the mean-value theorem, and mathematical induction. The definitions and results emphasized by the course are nevertheless stated and developed here.

A. — Derivatives and equations of evolution

Derivative and differentiability. Let $I \subseteq \mathbb{R}$ be an open interval, let $f: I \rightarrow \mathbb{R}$, and let $x_0 \in I$. The *derivative* of f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided this real limit exists. In that case, f is *differentiable at x_0* . It is differentiable on I if it is differentiable at every $x_0 \in I$. The right-hand and left-hand derivatives are defined by replacing $h \rightarrow 0$ with $h \rightarrow 0^+$ and $h \rightarrow 0^-$, respectively. Thus f is differentiable at x_0 precisely when both one-sided derivatives exist and are equal.

1)

- a) **DQPL** *Difference Quotient for a Power Law.* Define $p: \mathbb{R} \rightarrow \mathbb{R}$ by $p(x) = x^2$ for every $x \in \mathbb{R}$. Starting from the difference quotient, prove that $p'(x_0) = 2x_0$ for every $x_0 \in \mathbb{R}$.
- b) **ONDD** *One-sided Non-Differentiability Demonstration.* Define $a: \mathbb{R} \rightarrow \mathbb{R}$ by $a(x) = |x|$ for every $x \in \mathbb{R}$. Compute the right-hand and left-hand derivatives of a at 0 and prove that a is not differentiable there.
- c) **HDC** *Higher Derivative Conventions.* When f is differentiable, its derivative function is also written

$$\frac{df}{dx} = f'(x) = \frac{d}{dx} f(x).$$

If f' is differentiable, define $f'' = (f')'$ and write $d^2f/dx^2 = f''(x)$. Let $f: I \rightarrow \mathbb{R}$ be differentiable, put $J = \{t \in \mathbb{R} : 2t \in I\}$, and define $g: J \rightarrow \mathbb{R}$ by $g(t) = f(2t)$. Use the difference quotient to prove $g'(t) = 2f'(2t)$ for every $t \in J$.

Differential and difference equations. Let $I \subseteq \mathbb{R}$ be an interval. A *differential equation* is an equation imposed on an unknown function $x: I \rightarrow \mathbb{R}$ and some of its derivatives. A *solution on I* is a function possessing the required derivatives and satisfying the equation at every $t \in I$. When the argument is time t , write

$$\dot{x}(t) = \frac{dx}{dt}(t), \quad \ddot{x}(t) = \frac{d^2x}{dt^2}(t).$$

An *initial condition* prescribes one or more values of the unknown function and its derivatives at a specified time.

2) Let $m \in \mathbb{R}$ satisfy $m > 0$. In a one-dimensional Newtonian model, $t \in \mathbb{R}$ denotes time, a twice differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}$ gives the particle's position, and its force is the function $F_x: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_x(t) = m\ddot{x}(t) \quad (t \in \mathbb{R}).$$

- a) **PADE** *Prescribed Acceleration Differential Equation.* Suppose that $F_x(t) = t^2 - 1$ for every $t \in \mathbb{R}$. Find all possible position functions, or equivalently all twice differentiable functions $x: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$m\ddot{x}(t) = t^2 - 1 \quad (t \in \mathbb{R}).$$

Prove that they are exactly the functions

$$x(t) = \frac{t^4}{12m} - \frac{t^2}{2m} + c_1t + c_0 \quad (t \in \mathbb{R}),$$

where $c_0, c_1 \in \mathbb{R}$.

- b) **PDNDE** *Position-Dependent Nonlinear Differential Equation.* Suppose instead that $F_x(t) = x(t)^2 - 1$ for every $t \in \mathbb{R}$, so that

$$m\ddot{x}(t) = x(t)^2 - 1 \quad (t \in \mathbb{R}).$$

Explain why the occurrence of $x(t)^2$ makes this equation nonlinear. Determine all constant solutions $x: \mathbb{R} \rightarrow \mathbb{R}$.

- c) **ICSU** *Initial Conditions Select Uniquely.* Let $x_0, v_0 \in \mathbb{R}$. Among the solutions in part (a), find the unique one satisfying $x(0) = x_0$ and $\dot{x}(0) = v_0$.

Indication.— For the converse statement in part (a), subtract the displayed candidate from an arbitrary solution and use the standard consequence of the mean-value theorem that a function with zero second derivative is affine.

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. A real sequence is a map $F: \mathbb{N}_0 \rightarrow \mathbb{R}$, whose value at $n \in \mathbb{N}_0$ is denoted by F_n . A *difference equation* is a relation between terms of an unknown sequence. A solution is a sequence satisfying that relation for every index for which it is stated.

- 3) Consider the Fibonacci difference equation

$$F_{n+2} - F_{n+1} - F_n = 0 \quad (n \in \mathbb{N}_0).$$

- a) **FRCS** *Fibonacci Recurrence Computes Successors.* For the solution with $F_0 = 0$ and $F_1 = 1$, compute $F_2, F_3, \dots, F_7$.
- b) **TRIVFS** *Two Recurrence Initial Values Fix the Sequence.* Let $F, G: \mathbb{N}_0 \rightarrow \mathbb{R}$ be two solutions. Prove by induction that $F_0 = G_0$ and $F_1 = G_1$ imply $F_n = G_n$ for every $n \in \mathbb{N}_0$.
- c) **RANC** *Recurrence Alone is Not Complete.* Exhibit two distinct solutions of the difference equation. Explain why the recurrence without two initial values does not specify a unique sequence.

The explicit formula for the Fibonacci solution will be derived in Part VI. These first examples also illustrate a recurring method: one may guess a candidate solution, but the candidate becomes a solution only after its domain, differentiability, and governing equation have been checked.

B. — Small and bounded asymptotic orders

Asymptotic notation. Let $I \subseteq \mathbb{R}$ be an interval, let $x_0 \in \mathbb{R} \cup \{-\infty, +\infty\}$ be a limit point of I in the relevant direction, and let $f, g: I \rightarrow \mathbb{R}$. All limits below are taken through points $x \in I$. A *deleted neighbourhood* of a finite x_0 consists of the points satisfying $0 < |x - x_0| < \delta$ for some $\delta > 0$; at $+\infty$ it consists of the points satisfying $x > M$ for some $M \in \mathbb{R}$, and at $-\infty$ it is defined analogously.

Suppose first that $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 . We write

$$f(x) = o(g(x)) \quad \text{as } x \rightarrow x_0$$

if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

We write

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow x_0$$

if there exist $C \in \mathbb{R}$ with $C \geq 0$ and a deleted neighbourhood of x_0 throughout which

$$|f(x)| \leq C|g(x)|.$$

The latter definition remains meaningful without the non-vanishing assumption on g . Informally, $o(g)$ is negligible compared with g , whereas $O(g)$ is bounded in magnitude by a constant multiple of g .

4)

- a) **SROZ** *Square-Root Order near Zero*. On the interval $(0, \infty)$, prove that $x = o(\sqrt{x})$ as $x \rightarrow 0^+$.
- b) **SRGI** *Square Root Growth at Infinity*. On the interval $(0, \infty)$, prove that $\sqrt{x} = o(x)$ as $x \rightarrow +\infty$.
- c) **LSOB** *Local Sine Order Bound*. Using the elementary inequality $|\sin u| \leq |u|$ for every $u \in \mathbb{R}$, prove that $\sin(2x) = O(x)$ as $x \rightarrow 0$.
- d) **SBI** *Sine Bounded at Infinity*. Prove that $\sin(2x) = O(1)$ as $x \rightarrow +\infty$, although $\sin(2x)$ has no limit as $x \rightarrow +\infty$.
- e) **VRGBR** *Vanishing Ratio Gives a Bounded Ratio*. Let $f, g: I \rightarrow \mathbb{R}$ satisfy the hypotheses in the definition of small order. Prove that

$$f(x) = o(g(x)) \implies f(x) = O(g(x)) \quad (x \rightarrow x_0).$$

- f) **ONCRF** *Order Notation denotes a Class Rather than a Function*. Explain why $o(g)$ denotes a class of functions rather than one function. Interpret the conventional equality $f = o(g)$ as a membership assertion, and state that assertion using the symbol $\in$.

5) In every part, all functions are real-valued and all asymptotic statements refer to the same limit $x \rightarrow x_0$.

- a) **SRAC** *Scaling Rules for Asymptotic Classes*. Let $\lambda \in \mathbb{R}$. Prove that $f = o(g)$ implies $\lambda f = o(g)$, and that $f = O(g)$ implies $\lambda f = O(g)$.
- b) **SAR** *Sum of Asymptotic Remainders*. Prove that $f_1 = o(g)$ and $f_2 = o(g)$ imply $f_1 + f_2 = o(g)$. Prove the analogous assertion with O in place of o .
- c) **PROB** *Product Remainder Order Bound*. Prove that $f = O(g)$ and $p = O(q)$ imply $fp = O(gq)$. Deduce that, as $h \rightarrow 0$, the relations $r(h) = o(h)$ and $s(h) = O(h)$ imply $r(h)s(h) = o(h^2)$.
- d) **CRCL** *Composition Remainder Control Lemma*. Let $\delta > 0$, let $k: (-\delta, \delta) \rightarrow \mathbb{R}$ satisfy $k(h) \rightarrow 0$ and $k(h) = O(h)$ as $h \rightarrow 0$, and let ε be a real-valued function defined on an open interval containing 0, with $\varepsilon(u) \rightarrow 0$ as $u \rightarrow 0$. Prove

$$k(h)\varepsilon(k(h)) = o(h) \quad (h \rightarrow 0).$$

6) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$.

- a) **DILA** *Derivative Implies Linear Approximation*. Suppose that f is differentiable at x_0 . Starting from the derivative limit, prove

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h) \quad (h \rightarrow 0).$$

- b) **LADD** *Linear Approximation Determines the Derivative*. Conversely, suppose that $A \in \mathbb{R}$ and a function r , defined near 0, satisfy

$$f(x_0 + h) = f(x_0) + Ah + r(h), \quad r(h) = o(h) \quad (h \rightarrow 0).$$

Prove that f is differentiable at x_0 and $f'(x_0) = A$.

- c) **LAC** *Linear Approximation Coefficient*. Prove that at most one number $A \in \mathbb{R}$ can occur as the coefficient of h in an expansion of the form in part (b).
- d) **QEO** *Quadratic Error Order*. For $f(x) = x^2$ on $\mathbb{R}$, show directly that

$$f(x_0 + h) = f(x_0) + 2x_0h + h^2.$$

Prove that $h^2 = O(h^2)$ and $h^2 = o(h)$ as $h \rightarrow 0$.

C. — Product, composition, and repeated differentiation

7) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $u, v: I \rightarrow \mathbb{R}$ be differentiable at x_0 .

- a) **PRE** *Product Remainder Expansion*. Introduce functions r_u, r_v , defined near 0, such that $r_u(h) = o(h)$, $r_v(h) = o(h)$, and

$$u(x_0 + h) = u(x_0) + hu'(x_0) + r_u(h),$$

$$v(x_0 + h) = v(x_0) + hv'(x_0) + r_v(h).$$

Multiply these two identities.

- b) **CTRC** *Cross-Term Remainder Control*. Using Question 5, prove that all terms in the product from part (a), other than the constant term and the terms linear in h , have sum $o(h)$.
- c) **DPR** *Derivative Product Rule*. Apply Question 6 to prove that

$$(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0).$$

8) Let $I, J \subseteq \mathbb{R}$ be open intervals, let $g: I \rightarrow J$, let $F: J \rightarrow \mathbb{R}$, and let $x_0 \in I$. Suppose that g is differentiable at x_0 and F is differentiable at $y_0 = g(x_0)$.

- a) **ICI** *Inner Composition Increment*. For all sufficiently small $h \in \mathbb{R}$, put $k(h) = g(x_0 + h) - g(x_0)$. Prove that

$$k(h) = hg'(x_0) + o(h), \quad k(h) = O(h), \quad k(h) \rightarrow 0 \quad (h \rightarrow 0).$$

- b) **ORF** *Outer Remainder Function*. For every sufficiently small $s \in \mathbb{R}$ with $y_0 + s \in J$, define

$$\varepsilon(s) = \begin{cases} \frac{F(y_0 + s) - F(y_0) - F'(y_0)s}{s}, & s \neq 0, \\ 0, & s = 0. \end{cases}$$

Prove that $\varepsilon(s) \rightarrow 0$ as $s \rightarrow 0$.

- c) **CRB** *Composed Remainder Bound*. Prove

$$F(g(x_0 + h)) - F(g(x_0)) = F'(y_0)k(h) + k(h)\varepsilon(k(h)).$$

Use Question 5(d) to obtain a linear approximation in h .

- d) **DCR** *Derivative Chain Rule*. Conclude that $F \circ g$ is differentiable at x_0 and

$$(F \circ g)'(x_0) = F'(g(x_0))g'(x_0).$$

Repeated derivatives and binomial coefficients. For $n \in \mathbb{N}_0$, set $f^{(0)} = f$ and, whenever the required derivatives exist, define $f^{(n+1)} = (f^{(n)})'$. Define $0! = 1$ and $n! = 1 \cdot 2 \cdots n$ for $n \in \mathbb{N}$. If $n \in \mathbb{N}_0$ and $r \in \mathbb{N}_0$ satisfies $0 \leq r \leq n$, define

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

9) Let $I \subseteq \mathbb{R}$ be an open interval. The functions below are assumed to possess every derivative that occurs.

- a) **ZOPC** *Zeroth and Onefold Product Cases*. For $u, v: I \rightarrow \mathbb{R}$, verify that

$$(uv)^{(n)} = \sum_{r=0}^n \binom{n}{r} u^{(r)} v^{(n-r)} \quad (L_n)$$

holds for $n = 0$ and $n = 1$.

- b) **DIH** *Differentiated Induction Hypothesis*. Let $n \in \mathbb{N}$. Assume (L_n) and differentiate its two sides. Use the product rule to obtain two finite sums representing $(uv)^{(n+1)}$.

- c) **PCR** *Pascal Combination after Reindexing*. Reindex one of the sums from part (b), and prove directly from the factorial definition that

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r} \quad (1 \leq r \leq n).$$

Combine the two sums, treating the terms with $r = 0$ and $r = n + 1$ separately.

- d) **GLDFP** *General Leibniz Derivative Formula for Products*. Use induction to prove that, for every $n \in \mathbb{N}_0$ and all n -times differentiable functions $u, v: I \rightarrow \mathbb{R}$,

$$(uv)^{(n)}(x) = \sum_{r=0}^n \binom{n}{r} u^{(r)}(x) v^{(n-r)}(x) \quad (x \in I).$$

Indication.— In part (c), replace $r+1$ by a new index in the sum containing $u^{(r+1)}v^{(n-r)}$ before combining coefficients.

D. — Taylor polynomials and formal series

Taylor polynomial. Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives through order n at x_0 . The *Taylor polynomial of order n about x_0* is

$$T_{n,x_0}f(x) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} (x - x_0)^r \quad (x \in \mathbb{R}).$$

Admitted result. **TTPR** *Taylor Theorem with Peano Remainder*. Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$. Suppose that the derivatives $f^{(0)}, f^{(1)}, \dots, f^{(n)}$ exist on a neighbourhood of x_0 and that $f^{(n)}$ is continuous at x_0 . Then, as $h \rightarrow 0$ with $x_0 + h \in I$,

$$f(x_0 + h) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} h^r + o(h^n).$$

If, in addition, $f^{(n+1)}$ exists and is bounded on a neighbourhood of x_0 , the remainder after the displayed finite sum is $O(h^{n+1})$.

10) Retain the notation and hypotheses of the admitted Taylor theorem.

- a) **FTPC** *Finite Taylor Polynomial Construction*. Prove that, for every integer $k \in \mathbb{N}_0$ with $0 \leq k \leq n$,

$$(T_{n,x_0}f)^{(k)}(x_0) = f^{(k)}(x_0).$$

- b) **EVTF** *Equivalent Variable Taylor Form*. Put $x = x_0 + h$ and rewrite the admitted result as an expansion of $f(x)$ as $x \rightarrow x_0$. State the remainder in both the o form and, under the additional hypothesis, the O form.
- c) **LNAL** *Local Nature and Analytic Limitation*. Explain precisely why a statement as $x \rightarrow x_0$ controls f only near x_0 . Explain also why the admitted theorem, for one fixed finite $n \in \mathbb{N}_0$, does not assert that an infinite series converges to $f(x)$.
- d) **CTE** *Cubic Taylor Expansion*. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = x^3$. Compute $T_{3,1}f(x)$ and verify directly that it equals $f(x)$ for every $x \in \mathbb{R}$.

Taylor series. Let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives of every order at x_0 . The *Taylor series of f about x_0* is the formal power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

The Taylor series about $x_0 = 0$ is also called the *Maclaurin series*. The word *formal* means that this definition alone asserts neither convergence of the series nor equality of its sum with $f(x)$.

11)

- a) **PTSF** *Polynomial Taylor Series is Finite.* Let $m \in \mathbb{N}_0$, let $a_0, \dots, a_m \in \mathbb{R}$, and define $p: \mathbb{R} \rightarrow \mathbb{R}$ by

$$p(x) = \sum_{r=0}^m a_r x^r.$$

Prove that the Taylor series of p about 0 has no non-zero terms after degree m and is equal to $p(x)$ for every $x \in \mathbb{R}$.

- b) **FSD** *Formal Series Distinction.* State separately the following three assertions about a Taylor series: its coefficients are defined; the resulting numerical series converges at a specified $x \in I$; its sum at that x equals $f(x)$. Explain which assertion follows from the definition alone.
- c) **MTRF** *Maclaurin Terms for a Rational Function.* Define $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ by $f(x) = (1-x)^{-1}$. Prove that $f^{(n)}(0) = n!$ for every $n \in \mathbb{N}_0$ and hence determine its formal Maclaurin series.

E. — Quotients with vanishing numerator and denominator

An expression $f(x)/g(x)$ has the *indeterminate form* $0/0$ at $x_0 \in \mathbb{R}$ when $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow x_0$. The phrase does not assert that the quotient has a limit.

12) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f, g: I \rightarrow \mathbb{R}$ be differentiable at x_0 . Suppose $f(x_0) = g(x_0) = 0$ and $g'(x_0) \neq 0$.

- a) **FLQE** *First-order Linear Quotient Expansions.* Use Question 6 to write expansions of $f(x_0 + h)$ and $g(x_0 + h)$ to first order, with remainders in $o(h)$.
- b) **PNNZD** *Punctured-Neighbourhood Non-Zero Denominator.* Prove that there exists $\delta > 0$ such that $g(x_0 + h) \neq 0$ whenever $0 < |h| < \delta$.
- c) **LQDL** *Local Quotient Derivative Limit.* Divide the expansions from part (a) by h , justify the quotient using part (b), and prove

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{f'(x_0)}{g'(x_0)}.$$

Indication. — For part (b), use $g(x_0 + h)/h \rightarrow g'(x_0) \neq 0$ and the fact that a function with a non-zero limit is non-zero sufficiently near the limit point.

Admitted result. **GDQLR** *General Derivative Quotient Limit Rule.* Let $x_0 \in \mathbb{R}$, let $\delta > 0$, put $I = (x_0 - \delta, x_0 + \delta)$, and let $f, g: I \rightarrow \mathbb{R}$ be continuous. Suppose that f and g are differentiable at every $x \in I \setminus \{x_0\}$, that

$$f(x_0) = g(x_0) = 0, \quad g'(x) \neq 0 \quad (x \in I \setminus \{x_0\}),$$

and that there exists $L \in \mathbb{R}$ such that

$$\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L.$$

Then $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = L.$$

13)

- a) **SQR** *Sine Quotient Recovery.* Apply Question 12 to $f(x) = \sin x$ and $g(x) = x$ on $\mathbb{R}$ to recover

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

- b) **CSOQL** *Cosine Second-Order Quotient Limit.* Explain why Question 12 does not apply directly to $(1 - \cos x)/x^2$ at 0. Use the admitted general rule and part (a) to prove

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

- c) **RDQE** *Repeated Derivative Quotient Evaluation.* Use the admitted general rule and part (b) to prove

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}.$$

- d) **HRVA** *Hypothesis Review before Valid Application.* Define $f, g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = x$ for every $x \in \mathbb{R}$, $f(0) = 0$, and

$$f(x) = x^2 \sin(1/x) \quad (x \in \mathbb{R} \setminus \{0\}).$$

Prove directly that $f(x)/g(x) \rightarrow 0$ as $x \rightarrow 0$. Compute $f'(x)/g'(x)$ for $x \neq 0$ and show that this derivative quotient has no limit as $x \rightarrow 0$. Identify the failed hypothesis of the admitted general rule and explain why the original quotient may nevertheless converge.

End of Part I